INVESTOR EDITION • R8

TokenKickstarter Investor technical memo

TKS L1 + Cipher

Functional, performance & release evidence

Node v0.1.25 · Cipher CLI v0.2.3 · r8 evidence edition

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Classification: Investor technical memo

Test window: 2026-11-06 06:13:14 – 06:30:18

Evidence basis: Existing r8 captures; no new live tests were run for this edition.

> **Verified progress:** live RPC; all transfer bursts through 5,000 included; zero observed gas-limit overshoot; two-identity Cipher text and file delivery verified.

Three scope disclaimers

1. This is a project-prepared black-box technical assessment, not an independent assurance opinion, formal code-security audit, penetration test, financial audit or investment recommendation.
2. Findings describe a bounded test window and identified chain/release. They do not certify present uptime, future capacity, resilience or subsequent remediation.
3. Functional success is distinct from decentralization, governance execution and trustless verification. Material limitations are disclosed alongside results; untested or unproven claims remain **NOT VERIFIED**.

Read the executive summary and current limitations together.

INVESTOR OVERVIEW

Executive summary for investors

TokenKickstarter demonstrated a functioning public L1 and Cipher messaging path in the r8 window. The evidence supports concrete progress in transaction handling, RPC hardening, client messaging and release hygiene. It does not establish an independently operated validator network or unrestricted production reliability.

What the evidence demonstrates

Area	Verified observation
Chain identity	Chain ID 7779; node 0.1.25-8add90f713a; genesis 0x28a554…e31c
Normal-load latency	20/20 parallel transfers; inclusion p50 1.367 s; finality p50 4.009 s
Bursts	100 / 1,000 / 2,000 / 5,000 transfers all included
Gas accounting	Zero overshoots in burst sample and all three stress summaries
Cipher relay	1,000 stores succeeded; 1,000 acknowledgements; store rate about 277 messages/s
Cipher client	Two local identities; text both ways and matching text/binary file hashes
Release checks	Node and CLI release checksums matched; LICENSE, CI configuration and public bootnode present

What investors should also understand

The runtime remained **specVersion 412**, not 413. Two validators were operated on the same project VPS. The public RPC was pruned. WebSocket upgrades succeeded but CLI sessions reset; HTTPS mailbox delivery worked. A pending multisig item remained and successful 2-of-3 execution was not demonstrated.

All three 20,000-submission stress scripts completed, but that is **not** equivalent to all submissions being uniquely included. Peers reached zero; run 3 left one of ten senders short by 1,873 nonce increments. Anchor counters stayed at 37 anchors / 4,073 messages; trustless verification remained partial.

Overall reading: Verified functional progress with clearly identified operational and assurance work remaining. This memo provides technical diligence input, not a clean bill of health or a commercial-readiness certification.

Evidence: `r8/C3a_20parallel_finality.json`; `r8/C3_load_results.json`; `r8/C3s_post_nonce_final.json`; `r8/CIPHER_E2EE_summary.json`.

SCOPE & INTERPRETATION

How to read this memo

Evidence and status vocabulary

Label	Meaning in this report
VERIFIED / PASS	A specific captured observation or bounded test succeeded
IMPROVED IN r8	An earlier reported issue was absent, or a previously missing test was completed, in this window
PARTIAL / DEGRADED	Some functionality worked but a relevant adjacent path did not
NOT VERIFIED	Required evidence was absent or insufficient; not a claim of failure
OPEN	A material condition persisted in the captured deployment

A passing checksum test says that downloaded bytes matched a published checksum. It does not prove that the binary is free of defects. A successful file hash comparison demonstrates received content equality, not an independent cryptographic security proof. A published bootnode does not create independent validator operators.

Report navigation

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28–30	Methodology, evidence index and excerpt appendix

Evidence boundary

This edition rewrites the existing r8 assessment for investors. It does not repeat funded traffic, submit governance calls, inspect private infrastructure or assert that subsequent fixes have reached production. Operator documentation is distinguished from public endpoint observations. No private server access was used in the underlying stated method.

The project is the preparer named on the cover. The memo must not be represented as certification by an independent third-party audit firm.

VERIFIED PROGRESS

Verified progress versus earlier audits

The strongest progress claims are narrow and testable. “Improved” below means supported by r8 evidence, not permanently resolved across every workload or deployment.

Earlier issue or evidence gap	r8 observation	Appropriate investor wording
Gas-limit overshoot during prior stress	Zero in all three stress summaries; burst overshoot count zero	No gas overshoot observed in r8 load samples
Shard endpoint availability	/stats and /discovery returned 200	Previously problematic shard routes responded successfully
Cipher end-to-end test gap	CLI 0.2.3, two identities, text both ways and file hashes match	Client-to-client delivery verified in the tested setup
Release hygiene gaps	LICENSE and CI workflow captured	License and CI configuration present
Peer bootstrap discovery	Public bootnode file captured	Public bootstrap entry published
Larger stress evidence gap	Three 20,000-submission runs captured	Full stress evidence captured, including degradation and shortfall

Earlier benchmark context

Measure	r6 (prior-report context)	r7 (prior-report context)	r8
Inclusion / finality p50	1.35 / 4.62 s	1.37 / 5.03 s	1.37 / 4.01 s
Stress gas overshoots	3, up to about 1.9×	Not run	0 observed
Maximum stress block time	40 s	Not run	14 / 4 / 18 s

The historical columns reproduce the r8 report's comparison, rather than a new re-audit of earlier raw data. Workloads, client timing and network conditions may differ. They are directional context, not a controlled regression benchmark.

Not marked fixed: runtime deployment, shared validator host, pruning, WebSocket resets, unproven multisig execution, stress peer loss/shortfall and partial anchor verification. None is closed by the progress above.

Evidence: `r8/C3_gas_overshoot.json`; `r8/C3s_stress1.out`; `r8/C3s_stress2.out`; `r8/C3s_stress3.out`; `r8/CIPHER_E2EE_summary.json`.

CHAIN & SECURITY

Chain identity & deployed runtime

The identity tuple below anchors the scope of every performance and client result. The node version is a native executable version; the runtime version is an on-chain value. They are not interchangeable.

Field	Captured value	Interpretation
Network name	TKS Network	Public endpoint identification
Chain type	Live	Endpoint-reported classification
Chain ID	7779	EVM transaction domain
Native node version	0.1.25-8add90f713a	Matches the release version check
Runtime specVersion	**412**	Target 413 not demonstrated on-chain
Genesis	0x28a554…e31c	Full hash preserved in excerpt below

Selected identity fields


```
{
  "system_chain": "TKS Network",
  "system_version": "0.1.25-8add90f713a",
  "eth_chainId": "0x1e63",
  "specVersion": 412,
  "genesis": "0x28a554eb58356946b3ac7ce61fa1562cbda2dd1171ee8c7719140e0e33a2e31c"
}
```


This is a selected-field presentation of multiple RPC responses, not a single server response. The full genesis hash is public chain identity data, not a signing secret.

Investor relevance

A new binary release alone does not establish that an on-chain runtime upgrade occurred. Any subsequent claim that specVersion 413 is live should be supported by a fresh runtime query on the same canonical genesis and a documented upgrade event. This memo does not infer runtime remediation from repository tags or release announcements.

Evidence: `r8/A2_system_chain.json`; `r8/A2c_system_version.json`; `r8/A2d_system_chainType.json`; `r8/A3_runtime_version.json`; `r8/A4_eth_chainId.json`; `r8/A7_genesis_hash.json`.

CHAIN & SECURITY

Liveness, finality & validator topology

Public health at baseline


```
{
  "jsonrpc": "2.0",
  "id": 1,
  "result": {
    "peers": 2,
    "isSyncing": false,
    "shouldHavePeers": true
  }
}
```


The endpoint reported two peers and no active synchronization at the baseline check. Two block-number observations about a minute apart showed approximately one block per second in the low-load sample. These are liveness observations, not a month-long uptime measurement or a service-level guarantee.

Finality context

The idle monitor sampled best and finalized heads; the excerpt in the r8 source shows lag values of three to four blocks. The funded 20-parallel transaction test separately measured finality p50 at 4.009 seconds. Block production cadence must not be substituted for transaction finality latency.

Under stress, observed finality lag and block intervals increased. Maximum recorded lag reached 35 blocks in run 1 and 25 in run 3; peers reached zero in all three stress summaries. Baseline health therefore cannot stand in for stressed health.

Deployment concentration

Two Aura/GRANDPA authorities were captured. Project deployment documentation identifies both validators as running on the same TokenKickstarter VPS. This is a shared-host, single-operator deployment, not independently operated consensus infrastructure. The black-box work did not independently inspect physical servers or cloud tenancy.

Two shard-registry records also pointed to the same shard URL. Distinct validator accounts or advertised regions do not establish separate failure domains.

Conclusion: Chain liveness and transaction finality were observed. Multi-operator resilience, host-failure tolerance and long-duration availability remain outside what was demonstrated.

Evidence: `r8/A1_system_health.json`; `r8/A5a_eth_blockNumber.json`; `r8/A5b_eth_blockNumber.json`; `r8/F_finality_idle.json`; `r8/E1_chain_state.json`; `r8/doc_VALIDATORS.md`.

CHAIN & SECURITY

Public RPC security posture

Captured public-surface checks

Probe family	r8 observation	Bounded conclusion
author_insertKey / rotateKeys / hasKey	Unsafe calls rejected	Tested key-management methods not usable externally
localPeerId / listen / peers / networkState	Method not allowed	Tested node-identity/network disclosures blocked
eth_sendTransaction	No signer available	No unlocked signing path demonstrated by this probe
rpc_methods	152 methods listed	Method inventory captured; listing alone does not grant execution

Representative rejection responses

```
{
  "error": {
    "code": -32601,
    "message": "Method not allowed on public RPC"
  }
}
```

```
{
  "error": {
    "code": -32601,
    "message": "RPC call is unsafe to be called externally"
  }
}
```

The excerpts are selected error fields. Original response envelopes remain in the evidence files.

What can be said

The tested identity-disclosure and unsafe authoring calls were refused at the public endpoint. This is a useful reduction in exposed administrative capability. No privilege escalation or exploit was attempted, and privileged governance state was inspected read-only.

What cannot be inferred

These probes do not establish exhaustive RPC authorization coverage, rate-limit robustness, denial-of-service resistance, wallet safety, storage security or absence of hidden administrative interfaces. The report is not a vulnerability assessment of every exposed method. A rejected key-management RPC also does not prove historical key rotation or secure key custody.

Status: Public identity hiding and tested unsafe-method rejection verified within the sampled interface. Comprehensive security assurance **NOT VERIFIED**.

Evidence: `r8/B_insertKey.json`; `r8/B_rotateKeys.json`; `r8/B_hasKey.json`; `r8/B_localPeerId.json`; `r8/B_networkState.json`; `r8/B_eth_sendTransaction.json`; `r8/B_rpc_methods.json`.

CHAIN & SECURITY

Governance: sudo & multisig status

The captured chain includes Sudo and Multisig pallets. Published governance documentation describes the sudo account as derived from a 2-of-3 multisig. The presence of that configuration is different from evidence that two signers successfully exercised it in production.

Read-only findings

Item	Observation
Sudo account	0x8eeb…; roughly 1,000 TKS in the balance snapshot
Pending call identifier	0x1cb3…
Captured approvals	One
Pending timepoint	Height 14,119; extrinsic index 1
Successful 2-of-3 sudo execution	**NOT VERIFIED in this window**

Selected on-chain pending state


```
{
  "sudoAccount": "0x8eEb8be59CcB1C65211aB787e6BA174753A68f1A",
  "pendingCallHash": "0x1cb32578098d384761769705aad31c7316a324eff853d5cbb46c705d44cb50d7",
  "when": {
    "height": 14119,
    "index": 1
  },
  "approvalCount": 1
}
```


The earlier remediation tracker described the stale operation as cleared. The captured on-chain storage still contained it. This report gives the observed chain state precedence over that written status. The discrepancy is disclosed without implying malicious activity or a proven governance exploit.

Closure evidence required

A benign, preauthorized multisig exercise should demonstrate distinct approvals, successful inclusion and dispatch, expected state change, and a documented resolution or cancellation of the pending item. Extrinsic/block evidence is required; EVM receipts alone are not the right proof for a Substrate governance call.

This memo did not submit, approve, cancel or execute any governance action. No governance signing material is reproduced.

Evidence: `r8/E1_chain_state.json`; `r8/E_balances.json`; `r8/doc_SUDO-MULTISIG.public.md`; `r8/doc_R6-AUDIT-RESPONSE.md`.

CHAIN & SECURITY

Supply, registry & history access

Supply snapshot: context, not a financial audit

The chain-state capture reported total issuance of approximately **1,001,863,593.92454 TKS** across 37 accounts at that snapshot. The original r8 report noted that large holders remained unlabeled. These observations are not a cap-table reconciliation, ownership attestation, vesting audit or review of economic rights.

The supply snapshot is a read-only chain observation. It does not establish beneficial ownership, distribution fairness or control over the largest balances. No signing material is included in this investor edition.

Username registry

Check	Evidence result
Captured registered names	Seven
Calls exposed in metadata	registerUnsigned, registerBatchUnsigned, register, release, transfer
Read interfaces	names, reverseLookup, totalNames and palletVersion
New registration in this window	Not submitted / **NOT VERIFIED**

Registry metadata and existing entries demonstrate an available state surface. They do not demonstrate every lifecycle operation or prevent abusive registrations.

Public historical access

The pruning probe returned a block at head minus 200, but not at head minus 300. This establishes limited history availability at the sampled endpoint; it is not a complete measurement of an exact pruning policy.

Recent inclusion evidence should be captured promptly. Investors and independent reviewers cannot assume that old blocks, state or receipts will remain retrievable from this public RPC. An archive service or sufficiently long retention remains a useful diligence prerequisite.

Conclusion: Supply and registry state were readable, but ownership labeling, tokenomics and full historical replay are not established by this work.

Evidence: `r8/E1_chain_state.json`; `r8/E_balances.json`; `r8/F_name_registry.json`; `r8/ARCHIVE_prune_check.json`.

L1 PERFORMANCE

Performance: definitions & evidence quality

The L1 workload consists of funded EVM transfer tests, not a comprehensive sample of smart-contract execution. Cipher stores and acknowledgements are a different data-plane operation and are reported separately.

Terms used in the results

Term	Meaning / caution
Accepted	Submission RPC returned acceptance; does not by itself prove inclusion
Included	Receipt or poller observation; provenance differs by test
Wall TPS	Script-reported transaction count divided by its wall-time interval
Block TPS	Script-reported count over the selected block interval; a different denominator
Peak	Maximum observed transactions in one sampled block; not a per-second service guarantee
Inclusion latency	Submission-to-inclusion interval observed by the client
Finality latency	Submission-to-observed-finalized interval in the 20-parallel test
Gas overshoot	Sampled gasUsed exceeding gasLimit

Important stress measurement qualification

The stress poller reported **20,338** and **20,181** included transactions in runs that each submitted 20,000. Those numbers cannot be read as uniquely confirmed test transactions. The exact source of excess counts was not resolved in the captured evidence; overlapping observation, backlog or broader block counts are possible, not established explanations.

Accordingly, stress TPS in this memo is labeled **script-reported** and is not promoted as independently reconciled unique-transaction capacity. The per-sender nonce audit supplies a separate completion check and reveals a run-3 shortfall.

Interpretation rule

Use the 20-parallel test for its bounded latency observation, bursts for their reported all-included result, and stress for degradation characterization. Do not average unrelated metrics into a production capacity commitment. Acceptance speed and peak block counts are not equivalent to sustainable finalized throughput.

Evidence: `r8/C3a_20parallel_finality.json`; `r8/C3_load_results.json`; `r8/C3s_stress1.json`; `r8/C3s_stress2.json`; `r8/C3s_stress3.json`.

L1 PERFORMANCE

Parallel transfer latency: 20 of 20

Results

Measure	Captured result
Submitted / accepted / included	20 / 20 / 20
Successful / finalized	20 / 20
Inclusion minimum	0.731 s
Inclusion p50	**1.367 s**
Inclusion maximum	2.602 s
Finality minimum / p50	4.009 s / **4.009 s**
Finality maximum	5.463 s
Inclusion blocks	536156, 536157, 536158

Selected JSON fields


```
{
  "sent": 20,
  "accepted": 20,
  "included": 20,
  "success": 20,
  "finalized": 20,
  "inclusionMs": {
    "min": 731,
    "p50": 1367,
    "max": 2602
  },
  "finalityMs": {
    "min": 4009,
    "p50": 4009,
    "max": 5463
  }
}
```


Investor interpretation

The sample demonstrates successful simultaneous transfer handling with client-observed median inclusion of roughly 1.37 seconds and median finality of roughly 4.01 seconds. The distinction matters: a quickly produced block is not itself proof that a transaction has finalized.

Twenty observations are too few to characterize extreme percentiles, congested conditions or a long-term latency distribution. No p95 or p99 service objective is established here. Stress results later in this report show materially different operating conditions.

Evidence: `r8/C3a_20parallel_finality.json`; `r8/C3a_run.log`.

L1 PERFORMANCE

Transfer bursts: 100 through 5,000

Captured burst results

Burst size	Included	Wall TPS	Block TPS	Peak per block	Blocks
100	100 / 100	23.2	25.0	94	4
1,000	1,000 / 1,000	232.2	500.0	878	2
2,000	2,000 / 2,000	79.5	166.7	613	12
5,000	5,000 / 5,000	171.3	312.5	849	16

All four bursts were reported fully included. No gas overshoot was observed in the burst-block sample.

What these measurements show

The node processed the four specified short transfer bursts, including the largest 5,000-transfer case. Results vary substantially by run and timing. The smaller 1,000 burst had higher wall TPS than the larger 2,000 and 5,000 bursts, so no monotonic capacity curve can be inferred.

Wall-time and block-interval throughput answer different questions. The wall interval includes the client's timing boundaries; the block interval describes selected inclusion blocks. Reporting both is useful only when the denominator remains explicit. The higher of the two should not be singled out as the network's general capacity.

Boundaries of the conclusion

These were simple funded transfers on a project-operated network. Contract-heavy transactions, geographically distributed validators, sustained business traffic, failure recovery and adversarial workloads were not covered. “All included” is a result for these bursts, not a guarantee that every later larger stress run completed.

The stress section intentionally retains the run-3 shortfall. It would be misleading to use these successful bursts to dismiss that separate failure mode.

Evidence: `r8/C3_load_results.json`; `r8/C3_load_console.log`; `r8/C3_gas_overshoot.json`.

L1 PERFORMANCE

Gas accounting: no observed overshoot

Key verified improvement

The r8 burst sample and all three stress summaries recorded zero gas-limit overshoots. This is the clearest bounded improvement over the earlier r6 comparison, which reported three stress overshoots.

Check	r8 observation
Burst overshoot count	0
Burst blocks sampled	31
Stress run 1 overshoots	0
Stress run 2 overshoots	0
Stress run 3 overshoots	0
Sampled block gas limit	50 million gas

Selected burst gas-check fields


```
{
  "overshootCount": 0,
  "over": [],
  "sampleCount": 31
}
```


Meaning of the check

The check compares observed block gas usage against the block gas limit. Zero overshoot is a positive accounting observation in the examined blocks. It is not evidence that arbitrary future blocks, alternative transaction types or every consensus validation path have been exhaustively verified.

The configured gas limit is not itself measured throughput. Workload gas cost, scheduling, execution time, propagation and finality all affect capacity. This investor edition therefore emphasizes actual observed tests instead of extrapolating from a configuration number.

Appropriate remediation wording

“Gas overshoot observed in earlier stress testing was not observed in the r8 burst and stress samples.” That statement is supported. “Gas accounting is permanently fixed in every case” would go beyond the evidence.

Stress peer drops and the sender shortfall remained despite this improvement. Gas compliance and load resilience are separate acceptance dimensions.

Evidence: `r8/C3_gas_overshoot.json`; `r8/C3s_stress1.out`; `r8/C3s_stress2.out`; `r8/C3s_stress3.out`.

L1 PERFORMANCE

Stress overview: three 20,000-submission runs

Each run submitted 20,000 transfers across ten senders. All three scripts finished and reported 20,000 RPC acceptances with zero RPC submission errors. Completion of the script is not synonymous with complete transaction settlement.

Measure	Run 1	Run 2	Run 3
Submitted / accepted	20,000 / 20,000	20,000 / 20,000	20,000 / 20,000
Poller count	20,338	20,181	15,541
Wall TPS, script-reported	103.0	153.1	62.4
Block TPS, script-reported	303.6	325.5	239.1
Peak per block	988	942	899
Maximum block interval	14 s	4 s	18 s
Gas overshoots	**0**	**0**	**0**
Minimum observed peers	**0**	**0**	**0**
Maximum finality lag	35 blocks	4 blocks	25 blocks

Measurement caution

Run 1 and run 2 poller counts exceed the number submitted. This prevents treating those counters or derived rates as independently reconciled unique-load throughput. Run 3's poller count is below submissions, while the separate nonce audit shows a smaller cumulative shortfall. The two observation methods are not interchangeable.

Stress conclusion

The network accepted the submitted load and maintained sampled gas limits, but peer counts fell to zero and timing degraded. The nonce audit ended with nine senders at the expected nonce and one sender short by 1,873. This is an incomplete load outcome, not a clean stress pass.

No lasting outage, packet-level root cause, permanent transaction loss or eventual recovery beyond the captured window is proven. Those questions require follow-up evidence.

Evidence: `r8/C3s_stress1.json`; `r8/C3s_stress2.json`; `r8/C3s_stress3.json`; `r8/C3s_post_nonce_final.json`.

L1 PERFORMANCE

Stress runs 1 & 2: detailed observations

Run 1

Parameter	Captured value
Client start / end	2026-11-06 06:19:17.810 / 06:22:36.339
Submission interval / rate	25.874 s / 773.0 accepted submissions per second
Script wall interval	197.374 s
Selected block range	536331–536397 (67 blocks)
Median / maximum block interval	2 s / 14 s
Poller count / wall TPS	20,338 / 103.0, script-reported

The submission rate is an RPC-ingress measure. It must not be presented as finalized capacity. Run 1 showed the highest observed finality lag, 35 blocks, and a peer minimum of zero. Gas overshoot remained zero.

Run 2

Parameter	Captured value
Client start / end	2026-11-06 06:22:50.917 / 06:25:04.688
Submission interval / rate	26.142 s / 765.1 accepted submissions per second
Script wall interval	131.800 s
Selected block range	536413–536474 (62 blocks)
Median / maximum block interval	2 s / 4 s
Poller count / wall TPS	20,181 / 153.1, script-reported

Run 2 had the strongest script-reported wall throughput of the three stress attempts and lower observed maximum block interval. It still recorded zero peers at its minimum, so it is not evidence that network stability was resolved.

Common qualification: Both poller counts exceed 20,000. The excess is retained transparently, not relabeled as extra successful test transfers. Per-transaction reconciliation would be required before a unique-load throughput claim.

Evidence: `r8/C3s_stress1.json`; `r8/C3s_stress1.out`; `r8/C3s_stress2.json`; `r8/C3s_stress2.out`.

L1 PERFORMANCE

Stress run 3 & sender completion proof

Run 3 results

Parameter	Captured value
Client start / end	2026-11-06 06:25:19.233 / 06:29:48.613
Accepted submissions / RPC errors	20,000 / 0
Script wall interval	249.111 s
Selected block range	536493–536557 (65 blocks)
Poller count / wall TPS	15,541 / 62.4, script-reported
Median / maximum block interval	2 s / 18 s
Minimum peers / maximum finality lag	0 / 25 blocks
Gas overshoot	0

Cumulative nonce cross-check

The r8 completion calculation expected each sender to reach nonce **11,936**, based on the recorded starting nonce and test sequence. The final snapshot instead showed:

Sender group	Count	Final nonce	Difference from expected
Senders 1–9	9	11,936	0
Sender 10 (0xac49…6BA5)	1	10,063	**1,873 short**

The comparison is cumulative across the captured load sequence, not a per-receipt proof for every run. It supports the original r8 finding of a run-3 sender shortfall. It does not identify each missing hash or prove whether those transactions later landed outside the window.

The original report also states that the pool was empty after the tests. An empty pool plus acceptance responses is not equivalent to complete inclusion. Recovery, resubmission safety and exact failure cause remain unproven.

Investor disclosure: One of ten senders did not reach the expected ending nonce. This condition remains open and is not softened into a full completion claim.

Evidence: `r8/C3s_stress3.json`; `r8/C3s_post_nonce_final.json`; `r8/C1_senders_before.json`; `r8/C1_senders_after.json`.

L1 PERFORMANCE

Performance conclusions & reproducibility limits

What is established

- The bounded 20-parallel sample finalized all 20 transfers, with the measured latency distribution reported earlier.
- Bursts through 5,000 transfers were reported fully included.
- No gas-limit overshoot was observed in the sampled burst and stress evidence.
- The larger stress workload exposed degraded peer visibility, elevated block intervals and an incomplete sender outcome.

What is not established

Open measurement question	Why it matters
Unique test-hash count in each stress run	Poller counts exceed submissions in two runs
Exact attribution of throughput denominator	Wall interval, selected blocks and submission interval differ
Eventual status of short sender	Final nonce shortfall does not identify later recovery
Root cause of peers reaching zero	A monitor observation is not a packet or consensus diagnosis
Finality distribution under stress	The small latency test cannot substitute for stressed finality
Multi-host and contract-heavy behavior	Neither workload/topology was demonstrated

Recommended benchmark acceptance design

For a future approved test, retain a unique run identifier and every signed transaction hash, sender, nonce and submission response. Reconcile each hash to a canonical block and receipt, then to a finalized head. Publish timing definitions, retry policy and treatment of unrelated block traffic before comparing results.

Capture peer, head and finality monitors with sufficient history retention. Report missing, duplicate, rejected and reverted transactions separately, rather than using acceptance count as completion. Preserve any timeout as an explicit unresolved state until evidence changes it.

This is a proposed evidence standard, **not a new test performed by this edition**. No additional funds, load traffic or governance actions were used to prepare this investor memo.

Evidence: `r8/C3s_stress1.json`; `r8/C3s_stress2.json`; `r8/C3s_stress3.json`; `r8/C3s_post_nonce_final.json`.

CIPHER DATA PLANE

Cipher architecture & public endpoint matrix

Cipher was examined as a separate messaging/data plane: relay storage and acknowledgement, shard storage and discovery, signal/WebSocket behavior, and a two-identity client delivery test. These measurements are not interchangeable with L1 transaction throughput.

Endpoint observations

Path	Relay	Shard	Signal
/version	200	200	200
/health	405	200	405
/stats	200	200	200
/discovery	200	200	200
/hybrid/stats	200	200	200

The matrix reports the responses to the r8 probes. A 405 response means the probed HTTP method was not accepted at that route; it is not, alone, proof that the entire service was unavailable. Likewise a 200 response proves a successful route response, not correctness of every field or long-term service health.

Distinct roles and failure domains

Component	Role examined	Assurance boundary
Relay	Ciphertext store and acknowledgement; HTTPS mailbox	Availability and metadata still depend on infrastructure
Shard	Store path, discovery and anchor-related reporting	Readable stats are not proof of trustless verification
Signal / WSS	WebSocket upgrade and client-session behavior	Upgrade success did not imply stable sessions
CLI	Text and file delivery between two identities	Both clients were on one machine

Operator documentation describes the relay/shard production layout as a shared physical-host deployment. Two on-chain shard validator records pointed to the same public shard endpoint. Distinct service hostnames must not be described as proven infrastructure independence.

Evidence: `r8/relay_version.body`; `r8/shard_version.body`; `r8/signal_version.body`; `r8/doc_SHARD-AND-RELAY.public.md`; `r8/E_validators.json`.

CIPHER DATA PLANE

Relay: storage & acknowledgement results

1,000-message store exercise


```
{
  "n": 1000,
  "conc": 50,
  "ok": 1000,
  "fail": 0,
  "errs": {},
  "sec": 3.605,
  "rate": 277.4
}
```


The store script used 50 concurrent requests. It recorded 1,000 successes, no store failures and a duration of 3.605 seconds: approximately **277.4 successful store operations per second** in this sample.

Acknowledgement result


```
{
  "acked": 1000,
  "fail": 0
}
```


A separate acknowledgement pass recorded 1,000 acknowledgements with no failures. A smaller ten-message relay exercise was also captured.

Appropriate conclusion

The relay accepted the specified ciphertext-store workload and the acknowledgement phase succeeded. The quoted rate belongs to the store measurement. The evidence does not supply a single end-to-end store-plus-ack latency distribution, so it should not be advertised as a combined-path throughput service guarantee.

Storage acknowledgement does not by itself prove recipient decryption, durable replication, geographic redundancy, offline recovery or L1 anchoring. Those are different properties. Client delivery evidence and anchor limitations are therefore reported separately.

A short successful test does not establish retention duration or monthly availability. Those would require policy review, longer-duration retrieval and recovery tests, and independent deployment information.

Evidence: `r8/D2_relay_10msg.json`; `r8/D4_relay_1000msg.json`; `r8/D4_relay_1000_ack.json`.

CIPHER DATA PLANE

Shard & anchor verification status

Verified shard progress

The corrected-schema shard store exercise returned **10/10 successful stores**. The shard /stats and /discovery routes returned HTTP 200. Those route responses and bounded store success are verified improvements in r8.

Registry and anchor observations

Measure	Captured result
Registered swarm validators	Two
Advertised shard endpoint	Same URL for both records
Hyperswarm configuration	Eight shards; epoch 24
Anchor counter before / after	37 / 37
Anchored-message counter before / after	4,073 / 4,073
Challenge activity	Zero in the reported window
Trustless verification	**PARTIAL / NOT VERIFIED**

Anchor-counter response


```
{
  "jsonrpc": "2.0",
  "id": 1,
  "result": {
    "total_anchors": 37,
    "total_messages": 4073
  }
}
```


No increase in the displayed counters was observed in this window. Successful relay/shard stores cannot therefore be represented as evidence that these particular messages were anchored on L1. Unchanged counters alone do not prove the mechanism is permanently broken; scheduling and conditions outside the window were not established.

Trust and dispute model

The captured anchor documentation describes Merkle commitments with off-chain data availability and disputes involving fisherman/sudo. It states that automatic staking slashes for invalid anchors are not implemented in that documented model. No successful fraud challenge or automatic penalty was demonstrated here.

Conclusion: Store and discovery functionality worked. Complete trustless verification, challenge handling and independent availability assurance were not proven.

Evidence: `r8/D1_shard_10msg_ok.json`; `r8/shard_stats.body`; `r8/shard_discovery.body`; `r8/E_hyperswarm.json`; `r8/E_anchorStats_before.json`; `r8/E_anchorStats_final.json`; `r8/doc_ANCHOR-VERIFICATION.public.md`.

CIPHER DATA PLANE

WebSocket signalling: upgrade, then resets

Two different layers of evidence

Layer	Result	Meaning
Relay WebSocket upgrade	HTTP 101 Switching Protocols	Upgrade handshake accepted
Signal WebSocket upgrade	HTTP 101 Switching Protocols	Upgrade handshake accepted
CLI WebSocket sessions	Connection reset without closing handshake	Stable realtime path not demonstrated
HTTPS mailbox	Text/file delivery succeeded	Working alternative delivery path in this setup

Representative captured status

```
relay.tokenkickstarter.com /ws → 101 Switching Protocols
signal.tokenkickstarter.com /ws → 101 Switching Protocols
CLI send/recv/status → Connection reset without closing handshake
```

This is a concise presentation of the probe and CLI summary, not a complete packet transcript. The original captures retain the HTTP response headers and client logs.

Investor relevance

Realtime signalling remained degraded even while HTTPS-based mailbox operations worked. A successful upgrade is a connection setup event, not proof of a stable authenticated messaging session, reliable push delivery or reconnection recovery.

The r8 summary reports the reset on every CLI send/receive/status call in its sample. That observation should be treated as a functional limitation, not omitted because text and files eventually arrived through another path.

What remains to verify

A closure test should show a sustained client session, bidirectional live delivery, controlled reconnect behavior and no unexpected resets across a documented interval. Proxy, server and client causes should be distinguished with logs. This memo does not attribute the failure to any one vendor or component; the captured black-box evidence is insufficient for a root-cause conclusion.

Status: WSS degraded; HTTPS mailbox functional in the tested setup. No later fix is assumed.

Evidence: `r8/WSS_probe.json`; `r8/CIPHER_E2EE_summary.json`; `r8/send_a_to_b.log`; `r8/send_b_to_a.log`; `r8/wss_warn_counts.txt`.

CIPHER CLIENT EVIDENCE

Cipher CLI: two-identity text delivery

Setup and result

Item	r8 evidence
Client release	cipher-cli 0.2.3
Downloaded asset checksum	Match recorded
Identities	Two independent client identities on the same machine
Identity A → B	PASS; second send received
Identity B → A	PASS; both sends received
Client encryption status	AES-256-GCM · Double Ratchet
Delivery context	HTTPS mailbox functional; WSS resets recorded

Selected summary fields


```
{
  "version": "cipher-cli 0.2.3",
  "sha256_asset_match": true,
  "text_A_to_B": "PASS (second send received)",
  "text_B_to_A": "PASS (both sends received)",
  "encryption": "AES-256-GCM · Double Ratchet (CLI status)"
}
```


What was newly demonstrated

The test moved beyond raw relay-store success to observed delivery between two actual CLI identities, including both text directions. This closes an earlier functional evidence gap for the specific client version and setup.

The summary explicitly notes that A-to-B passed on the second send. It is not a claim that every first attempt was delivered without retry. WebSocket reset warnings also remained present. Successful delivery and transport reliability must be reported separately.

Assurance boundaries

Both identities were on the same machine. This was not a two-country or independent-host interoperability exercise. The reported cipher suite comes from CLI status; this memo does not independently prove cryptographic implementation correctness, forward secrecy, identity authentication or resistance to a malicious relay.

Status: Functional client-to-client text delivery verified in both directions, with transport qualifications retained.

Evidence: `r8/CIPHER_E2EE_summary.json`; `r8/send_a_to_b.log`; `r8/send_b_to_a.log`; `r8/recv_a.log`; `r8/recv_b.log`.

CIPHER CLIENT EVIDENCE

Cipher CLI: file integrity & confidentiality boundary

Verified file delivery

The client exercise transferred a text file and a **64 KiB binary file**. The received files matched their originals by SHA256. This is a direct integrity check of the recovered contents in the tested client workflow.

File	Source / received SHA256	Result
Text payload	Same digest below	PASS
64 KiB binary payload	Same digest below	PASS

Text payload digest

```
e53f10c55c7d0c81e05442fcac701c259d4751189dd63813a82e180cdf87e2d3
```

Binary payload digest

```
dcc04adc9670668ec5ff513d008b704202c53e254f2d36d286c331dfaf2c6be3
```

The digests are public test-file identifiers, not private keys. Full source/received path comparisons are retained in the evidence file.

Confidentiality statement: carefully bounded

The CLI reported AES-256-GCM and Double Ratchet, and the tested workflow delivered matching plaintext after client processing. This supports the reported E2EE functional workflow. It is not an independent review of protocol design, key storage, identity verification, attachment encryption or implementation security.

Availability and metadata exposure remain dependent on operator relays, shared infrastructure and the network edge. End-to-end encryption should not be equated with metadata anonymity, independent infrastructure or guaranteed delivery.

Not tested here

Large-attachment scaling, malicious attachment handling, interrupted-transfer recovery, cross-platform clients, multi-device synchronization and long-term offline retrieval were not established. The two tested payloads should not be generalized into an unrestricted file-transfer guarantee.

Evidence: `r8/CIPHER_file_sha_compare.txt`; `r8/CIPHER_E2EE_summary.json`; `r8/sendfile_a.log`; `r8/sendfile_bin_a.log`; `r8/recv_b_file.log`.

RELEASE & OPERATIONS

Release integrity: node & CLI

Version alignment and checksum observations

Artifact	Result
Node release tag	v0.1.25
Native node version	0.1.25-8add90f713a
Tag / live version alignment	Recorded as matching in r8
Node linux-amd64 checksum	Matches published SHA256SUMS
Cipher CLI version	0.2.3
Cipher linux-x64 release asset checksum	Match recorded

Node binary SHA256

```
3126b7a87313169d708c7cf41c4b21872430e2e0db1c53957827c5e67b7b1721
```

Cipher CLI linux-x64 archive SHA256

```
5f9576f2ac6be28be80b8c8d0e97243e3865632e3cdb2f9bd83424cf1d623576
```

Why this matters

The checksum checks reduce uncertainty about which published artifacts were used. The native node version aligns with the captured endpoint version. The client test is tied to an identified CLI release rather than an unspecified development build.

Limits of checksum assurance

A checksum hosted alongside a release is an integrity reference, not independent authentication of the publisher or reproducibility from source. No signed provenance chain, reproducible build, dependency-vulnerability audit or complete source review is established by these matches.

The runtime remained specVersion 412 despite the newer native node release. Artifact integrity and on-chain runtime deployment are separate statements and are kept separate throughout this report.

Evidence: `r8/G2_binary_sha256_local.txt`; `r8/G2_binary_version.txt`; `r8/G_SHA256SUMS_v0.1.25.txt`; `r8/G_tag_object.json`; `r8/G_tag_v0.1.25.json`; `r8/sha_local.txt`; `r8/CIPHER_E2EE_summary.json`.

RELEASE & OPERATIONS

Engineering hygiene & bootstrap readiness

Published engineering artifacts

Artifact	Captured observation	What it does not establish
LICENSE	License file present	Legal ownership review or compliance of all dependencies
CI workflow	Formatting, selected lint and unit-test commands configured	A passing current run, complete coverage or audited security
Public bootnode	Public P2P bootstrap entry published	Independent operator participation or guaranteed peer stability
Deployment notes	Validator, archive and stability documents captured	Verified completion of every operator commitment

CI scope in the captured workflow

The workflow targets Ubuntu 22.04, installs a stable Rust toolchain including rustfmt and clippy, and includes the wasm32 target. It configures:

- `cargo fmt --all -- --check`
- Clippy for cipher-merkle, cipher-shard-executor, cipher-hybrid and pallet-adaptive-gas with warnings denied.
- Unit tests for the same selected package set.

These are meaningful hygiene controls. The memo does not claim that the workflow ran successfully for the tested release, that every workspace package is covered, or that integration and adversarial tests are complete.

Bootstrapping versus decentralization

Publishing a bootnode helps independent nodes discover peers. It is a prerequisite-style operational improvement, not proof that independent nodes have joined consensus. The stress monitor still observed peers reaching zero, and the documented validator deployment remained two validators on one project VPS.

Stability-policy context

Captured project documentation describes historical genesis changes and future upgrade/reset policies. Those are operator statements. This report identifies the observed genesis and recommends preserving that identity in future evidence. It does not certify compliance with a future notice period or guarantee no resets.

Status: LICENSE, CI configuration and public bootnode present; execution effectiveness and independent operations remain separately verifiable work.

Evidence: `r8/doc_LICENSE`; `r8/doc_ci.yml`; `r8/doc_bootnodes.public.txt`; `r8/doc_MAINNET-STABILITY.md`; `r8/doc_VALIDATORS.md`.

CURRENT LIMITATIONS

Current limitations & roadmap disclosures

These conditions remained material in r8. No later remediation is assumed. They should accompany the verified strengths in any investor discussion based on this memo.

#	Current limitation	Investor-relevant implication
1	Runtime specVersion **412**; 413 not on-chain	Native release progress does not demonstrate runtime deployment
2	**Two validators on the same operator VPS**	Concentrated control and shared infrastructure failure domain; not multi-operator decentralization
3	**Pruned public RPC**; head−200 present, head−300 absent in probe	Independent historical replay constrained
4	**WSS resets after 101 upgrade**; HTTPS mailbox works	Realtime client transport remains degraded
5	**Pending multisig 0x1cb3…**, one approval; no proven 2-of-3 execution this window	Operational governance exercise and pending-state resolution remain unproven
6	**Stress peers reached zero**; one sender 1,873 nonce increments short; maximum block intervals 14 / 4 / 18 s	Acceptance did not assure full load completion or stable performance
7	**Anchors stayed 37 / 4,073**; zero challenges; trustless verification partial	Store success is not demonstrated end-to-end anchoring or automatic dispute enforcement

Additional diligence boundaries

Stress counters exceeded submitted counts in two runs, so derived stress rates are script-reported rather than uniquely reconciled capacity. Large supply holders were not fully labeled. Source-code security, legal compliance, economic design, custody controls, disaster recovery and long-duration service objectives were not audited.

Disclosure discipline

The report can reasonably say that functionality was demonstrated and selected issues improved. It should not say that all audit findings are closed, that infrastructure is decentralized, that multisig control has been exercised successfully, or that anchor trust is fully verified. The distinction is evidential, not a prediction about the project's eventual roadmap.

ROADMAP & DILIGENCE

Proposed remediation evidence & acceptance gates

The following is a recommended diligence roadmap, not a statement that work is scheduled, funded or completed. Any funded load, governance write or infrastructure change needs its own authorization.

Workstream	Evidence required to close the corresponding disclosure
Runtime deployment	Same-genesis runtime response at intended specVersion; upgrade block/extrinsic and post-upgrade functional checks
Independent operations	Separate operator/host documentation, distinct failure domains and observed validator participation; onboarding and recovery exercise
Historical replay	Archive or documented long-retention RPC; independent retrieval of previously captured blocks and state
WebSocket reliability	Sustained authenticated CLI sessions, bidirectional realtime delivery and controlled reconnects without unexplained resets
Governance execution	Benign preauthorized 2-of-3 action, distinct signer evidence, successful dispatch/state change and resolved stale operation
Stress reliability	Hash-by-hash finalized reconciliation, every sender at expected nonce, stable peers and disclosed timing/timeout criteria
Anchors and challenges	Test-message membership proof through off-chain data to on-chain root; exercised dispute path and clearly documented enforcement model

Suggested order of diligence

First establish reliable history and measurement reconciliation, so later claims can be independently replayed. Then verify runtime/governance state and realtime transport behavior. Scale topology and load only with instrumentation adequate to distinguish acceptance, inclusion, finality and recovery.

Closure rule

An issue moves from OPEN or NOT VERIFIED only when the specific acceptance evidence is captured on the identified deployment. A code change, tracker update, published target or successful adjacent component is not enough by itself.

No deadlines or delivery commitments are implied by this recommended order. The memo intentionally distinguishes verified progress from future work.

APPENDIX • METHOD

Appendix A — methodology & preservation

Source basis

This report was prepared from the existing r8 Markdown/PDF and selected raw evidence in `audit-20261106b/r8/`. The originals were left untouched. The investor edition is stored separately in `audit-20261106c/`.

Source test window: 2026-11-06 06:13:14 – 06:30:18.

The cover timestamp is the historical evidence window, not a fresh health attestation. No new RPC, relay, shard or load requests were needed to compile this edition. Selected stress timestamps were converted from the recorded machine timestamps into the same clock convention as the cover; original captures retain their native timestamp syntax.

Public targets in the r8 scope

- RPC: `https://rpc.tokenkickstarter.com`
- Relay: `https://relay.tokenkickstarter.com`
- Shard: `https://shard.tokenkickstarter.com`
- Signal: `https://signal.tokenkickstarter.com`

Captured source context: TokenKickstarter/tokenkickstarter-node at v0.1.25 and TokenKickstarter/Cipher-cli at v0.2.3. Repository/release references are historical evidence, not a fresh source review.

Underlying stated test method

Public endpoint black-box checks, published artifact checks, funded transfer workloads, relay/shard stores, WebSocket upgrade probes and two-identity CLI messaging. No server login, host inspection, exploit attempt or privileged governance write was part of the stated method. Sudo/multisig state was read-only.

Editorial controls in this edition

- Retain successful results and known limitations together.
- Attribute operator deployment statements to the captured documentation.
- Distinguish configured features, observed behavior and unverified properties.
- Use selected JSON fields where full dumps would distract from interpretation.
- Exclude signing material, credential files and unnecessary client identity details.
- Keep script-reported stress counters visible, including inconsistencies.

Reproduction and confidentiality

Evidence paths are local bundle references, not public download links. A reviewer needs a sanitized evidence package to follow them. Raw directories contain operational helpers and client working files and should not be distributed wholesale. The investor PDF and Markdown are the intended deliverables; nothing was sent or uploaded by this preparation step.

A source-preservation hash baseline and a post-build validation summary accompany the new files. Output-only scans are bounded checks, not a certification of every raw file's safety for external release.

APPENDIX • EVIDENCE INDEX

Appendix B — evidence index: chain & performance

All paths below are relative to the original `audit-20261106b/` directory. The `r8/` evidence remains in place. File references do not imply that every captured file is suitable for unrestricted redistribution.

Subject	Principal evidence paths
Baseline health	`r8/A1_system_health.json`
Network / native version	`r8/A2_system_chain.json`; `r8/A2c_system_version.json`
Runtime / EVM identity	`r8/A3_runtime_version.json`; `r8/A4_eth_chainId.json`
Genesis / head advance	`r8/A7_genesis_hash.json`; `r8/A5a_eth_blockNumber.json`; `r8/A5b_eth_blockNumber.json`
Idle finality	`r8/F_finality_idle.json`
RPC refusals	`r8/B_insertKey.json`; `r8/B_rotateKeys.json`; `r8/B_hasKey.json`; `r8/B_networkState.json`
RPC inventory / signer	`r8/B_rpc_methods.json`; `r8/B_eth_sendTransaction.json`
State / pending multisig / supply	`r8/E1_chain_state.json`; `r8/E_balances.json`
Validator topology statement	`r8/doc_VALIDATORS.md`
Pruning probe	`r8/ARCHIVE_prune_check.json`
Registry	`r8/F_name_registry.json`
Parallel latency	`r8/C3a_20parallel_finality.json`; `r8/C3a_run.log`
Bursts	`r8/C3_load_results.json`; `r8/C3_load_console.log`
Gas checks	`r8/C3_gas_overshoot.json`
Stress counters / monitors	`r8/C3s_stress1.json`; `r8/C3s_stress2.json`; `r8/C3s_stress3.json`
Stress concise summaries	`r8/C3s_stress1.out`; `r8/C3s_stress2.out`; `r8/C3s_stress3.out`
Nonce completion	`r8/C3s_post_nonce_final.json`; `r8/C1_senders_before.json`; `r8/C1_senders_after.json`

Reviewer priorities

Start with the identity tuple, then inspect the all-included burst results and the stress nonce snapshot. Treat the poller inconsistency as an unresolved measurement issue. Review the pending multisig storage and operator topology documentation before treating runtime or decentralization work as complete.

APPENDIX • EVIDENCE INDEX

Appendix C — Cipher / release index & closing note

Subject	Principal evidence paths
Relay store / ack	r8/D4_relay_1000msg.json; r8/D4_relay_1000_ack.json
Shard store	r8/D1_shard_10msg_ok.json
Shard routes	r8/shard_stats.body; r8/shard_discovery.body
WebSocket probe	r8/WSS_probe.json
Client result summary	r8/CIPHER_E2EE_summary.json
Bidirectional text	r8/send_a_to_b.log; r8/send_b_to_a.log; r8/recv_a.log; r8/recv_b.log
File delivery / integrity	r8/sendfile_a.log; r8/sendfile_bin_a.log; r8/CIPHER_file_sha_compare.txt
Anchor snapshots	r8/E_anchorStats_before.json; r8/E_anchorStats_final.json
Anchor / deployment docs	r8/doc_ANCHOR-VERIFICATION.public.md; r8/doc_SHARD-AND-RELAY.public.md
Node checksum / version	r8/G2_binary_sha256_local.txt; r8/G2_binary_version.txt; r8/G_SHA256SUMS_v0.1.25.txt
CLI checksum	r8/sha_local.txt; r8/CIPHER_E2EE_summary.json
Tag evidence	r8/G_tag_object.json; r8/G_tag_v0.1.25.json
License / CI / bootnode	r8/doc_LICENSE; r8/doc_ci.yml; r8/doc_bootnodes.public.txt

Selected anchor and acknowledgement fields


```
{
  "anchor_snapshot": {
    "total_anchors": 37,
    "total_messages": 4073
  },
  "relay_ack": {
    "acked": 1000,
    "fail": 0
  }
}
```


Closing assessment

The r8 evidence shows meaningful functional progress: live chain access, successful bounded transfer bursts, no observed sampled gas overshoot, working relay/shard paths and two-identity Cipher delivery including matching files. It also shows a concentrated deployment, incomplete stress completion, degraded realtime transport and unresolved runtime, governance and anchor-assurance evidence.

Investor technical memo · TokenKickstarter audit team · Bern, Switzerland. Read as a bounded project-prepared assessment, with current limitations preserved.